431 Quiz 2 for Fall 2020

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Instructions

Please visit the Quiz 2 web page for complete instructions at https://github.com/THOMASELOVE/431-2020/blob/master/quizzes/quiz2

This PDF contains 19 pages. Be sure you have all 19.

The Google Form Answer Sheet

Place your responses in the Google Form located at https://bit.ly/431-2020-quiz2-answer-sheet. All of your answers must be submitted through that Google Form by noon on Monday 2020-12-14, without exception. The form will close at that time, and no extensions will be made available, so do not wait until Monday to submit. We will not accept any responses except through the Google Form.

The Google Form contains places to provide your responses to each of the 31 questions (including a place to submit a file with your response to the essay in question 30), and a final affirmation where you'll type in your name to tell us that you followed the rules for the Quiz. You must complete that affirmation and then submit your results. When you submit your results (in the same way you submit a Minute Paper) you will receive an email copy of your submission, with a link that will allow you to edit your results.

If you wish to work on some of the quiz and then return later, you can do this by [1] typing in your name at the start of the Quiz, then [2] completing the affirmation (after Question 31) which asks you to type in your full name again (this time to affirm that you've followed these instructions), and then [3] submitting the quiz. You will then receive a link at your CWRU email which will allow you to return to the quiz as often as you like without losing your progress.

Timing

The Quiz is meant to take 5-6 hours. I expect most students will take 4-8 hours, and some will take as little as 3 or as many as 12. It is a bad idea to spend a long time on any one question without asking for help.

Responding to the Items

The Quiz contains 31 questions. The first 30 comprise the actual Quiz and are worth a combined 100 points. It is to your advantage to answer all 31 Questions. Your score is based on the number of correct responses, so there's no chance a blank response will be correct, and a guess might be, so you should definitely answer all of the questions.

26 of the first 29 questions are worth 3 points each, and the others (questions 16, 17, 19) are worth 4-6 points, as indicated.

- The first 15 questions all refer to the same data set and each have two parts to them. They test your ability to use R to analyze data, and are not arranged in order of difficulty.
- Questions 16-29 are not in any special order.
- Questions 1-29 range in difficulty from "things I expect everyone to get right" to "things that are deliberately tricky".
- In Questions 1-29, when you have the option to type in a response, just provide the answers, without adding in material to create a complete sentence.
  - So, for example, if you are asked how many students meet a particular standard and the answer is 10, please just type **10** into the form, as opposed to something like "There are 10 students."
  - Occasionally, we ask you to provide a single line of code. Do not include the `library` command at any time as part of your response. Assume in all questions that all relevant packages have been loaded in R.

Question 30 is an essay, which will require more time than any of the other questions. Question 30 requires the careful writing of complete sentences, which you will upload to the Google Form as either a PDF or Word document. You needn't use R Markdown to write your essay, although you are welcome to do so. Question 30 is worth 8 points.

Question 31 is about whether you've completed the class evaluation at https://webapps.case.edu/courseevals/. Question 31 has no impact on your Quiz 2 grade, but a response of `a` or `b` will add 1 point to your class participation grade.

The Data Set and R Scripts

I have provided one data set (called `survA.csv`) and two R scripts (our usual `Love-boost.R` and also `wc_code.R`) for this Quiz. They may be helpful to you. You will find them in our Shared Google Drive in the data subfolder within the Quiz 2 folder.

Getting Help

This is an open book, open notes quiz. You are welcome to consult the materials provided on the course website and that we've been reading in the class, but you are not allowed to discuss the questions on this quiz with **anyone** other than Professor Love and the teaching assistants. You will be required to complete a short affirmation that you have obeyed these rules as part of submitting the Quiz.

If you need clarification on a Quiz question, you have exactly two ways of getting help:

1. You can ask your question in a private post on Piazza to the instructors.
2. You can ask your question via email to **431-help at case dot edu**.

During the Quiz period (from now through Noon 2020-12-14) we will not answer questions about the Quiz except through the two approaches listed above. We promise to respond to all questions received before 9 AM on 2020-12-14 in a timely fashion.

A few cautions:

- Specific questions are more likely to get helpful answers.
- We will not review your code or your English for you.
- We will not tell you if your answer is correct, or if it is complete.
- Dr. Love will email all students and post a note to Piazza if we find an error in the Quiz that needs fixing.

When Should I ask for help?

We recommend the following process.

- If you encounter a tough question, skip it, and build up your confidence by tackling other questions.
- When you return to the tough question, spend no more than 10-15 minutes on it. If you still don't have it, take a break (not just to do other questions) but an actual break.
- When you return to the question, it may be much clearer to you. If so, great. If not, spend 5-10 minutes on it, at most, and if you are still stuck, ask us for help.
- This is not to say that you cannot ask us sooner than this, but you should **never, ever** spend more than 20 minutes on any question without asking for help.

R Packages Dr. Love Used To Build This

This **doesn't** mean you need to use all of these packages, but Dr. Love was able to build a complete answer sketch using this setup (although he doesn't promise to have used all of these packages.)

```
library(knitr)
library(janitor)
library(mosaic)
library(naniar)
library(ggrepel)
library(car)
library(magrittr)
library(patchwork)
library(vcd)
library(Epi)
library(Hmisc)
library(broom)
library(tidyverse)

opts_chunk$set(comment = NA)
theme_set(theme_bw())

source("data/Love-boost.R")

survA <- read_csv("data/survA.csv")
```

I've also provided my session information at the end of this PDF in case that helps you.

Good luck!

Setup for Questions 1-15

Questions 1-15 all refer to the same data set (the `survA.csv` data) which I have provided to you in our Shared Google Drive in the data subfolder within the Quiz 2 folder.

Consider the `survA.csv` file Dr. Love has provided. The file contains (lightly edited) responses to nine items that I have asked of students during the first week of 431, in each of the past seven years. The `survA.csv` file also includes a subject identification code called `student`, as well as the `year` (which ranges from 2014 through 2020) in which the survey was given. Ingest the data into R and complete analyses to address Questions 1-15.

- Note that each of the first 15 questions contains two parts.

1 Question 1

The `height_in` variable stores responses to the item "what is your height?" that have been converted to measurements in inches.

a. How many of the students who responded to the `height_in` item gave responses that fell within two standard deviations of the sample mean?
b. What percentage of students with a response to `height_in` would we expect to see in your count in Question 1a if the Normal distribution was a perfect model for the distribution? Round your answer to zero decimal places.

2 Question 2

The `lastsleep` variable contains responses to the item "How many hours did you sleep last night?"

a. In which `year` did the sample of `lastsleep` values have the smallest mean number of hours?
b. After rounding to one decimal place, what was the sample mean of `lastsleep` in the `year` you identified in Question 2a?

3 Question 3

The most common response to the item "What is your favorite color?", which was gathered in the `favcolor` variable was blue, mentioned by nearly half of the respondents. The second most common response was green.

a. What color was the third most common choice, and how many subjects gave that response?
b. What color was the fourth most common choice, and how many subjects gave that response?

4 Question 4

All subjects in the `survA` file were assigned a `student` code and `year` value by Dr. Love. The other nine variables were collected from their responses to the items on the survey.

a. Which of the nine variables (excluding `student` and `year`) provided has the largest number of missing values?
b. How many values are missing in the variable you identified in Question 4a?

5 Question 5

a. Identify the `student` code for the subject with the largest number of missing values.
b. For which variables (excluding `student` and `year`) has the subject you identified in Question 5a provided data?

6 Question 6

Ava and Ivy are two analysts working separately, who wish to compare the `haircut` prices across the survey years using an appropriate overall test of whether the average (mean or pseudo-median) `haircut` values are detectably different by year, using a 90% confidence level. Each analyst assumes all missing `haircut` prices are missing completely at random. Ava also assumes that each year's `haircut` prices follow a Normal distribution with similar variance across years. Ivy, on the other hand, assumes that the `haircut` prices are symmetric, but that they do not follow a Normal distribution due to problems in the tails of the distributions.

a. State the p values (rounded to three decimal places) for each of the two analysts (Ava and Ivy) under the assumptions stated above.
b. Which of the analysts will declare a statistically detectable difference between the average `haircut` prices across years?
    a. Ava alone
    b. Ivy alone
    c. Both Ava and Ivy
    d. Neither Ava nor Ivy

7 Question 7

The `lecture` variable describes agreement (1 = Strongly Disagree to 5 = Strongly Agree) with the statement "I prefer to learn from lectures over learning from activities." The `alone` variable described agreement (on the same 1-5 scale) with the statement "I prefer to work on projects alone over working on a team.") Create a 5x5 table comparing lecture to alone (with `lecture` in the rows and `alone` in the columns) for the subjects in the data who have complete information on the `lecture` and `alone` variables, and also have a value in the `height_in` variable.

a. Following the Question 7 instructions to build a table, you should find three cells which contain just one subject. Identify the `student` code for the tallest of those three subjects.
b. How many of the three subjects who are the only members of their cell in the table from Question 7a agreed more with the `lecture` statement than with the `alone` statement?

8 Question 8

Now use R to combine the responses 1 and 2 and 4 and 5 for the `lecture` and `alone` variables discussed in Question 7, and then create a new contingency table with `lecture` in the rows and `alone` in the columns. Your new table should have 3 rows and 3 columns. Now, use an appropriate hypothesis test to assess, with 90% confidence, whether the rows in your new table are independent of the columns.

a. What type of hypothesis test did you use to address this question?
    a. A test based on the t distribution
    b. A test based on the chi-square distribution
    c. A test based on a bootstrap
    d. A test based on the F distribution
    e. None of these
b. After rounding to five decimal places, what p value do you obtain from your test in Question 8a?

9 Question 9

Now consider the `english` variable, which specifies for each subject whether their most comfortable language is English, or some other language, and the `smoker` variable, which divides subjects into group 1 (non-smokers), or group 2 (former smoker), or group 3 (current smoker). For Question 9, including only the subjects with complete data on these two variables. Collapse the `smoker` responses of 2 and 3 into a single group, which we'll call the smokers for this question. Now, estimate the odds ratio for being a non-smoker for students who are most comfortable speaking in a language other than English, as compared to students who are most comfortable speaking in English.

a. Provide a point estimate for the odds ratio identified in Question 9, rounded to two decimal places.
b. Again rounding to two decimal places, specify a 90% confidence interval for the estimate you provided in Question 9a.

10 Question 10

There are 3 subjects in the `survA` data who are missing exactly two values across the entire data set.

a. Of the 3 subjects missing exactly two values, how many are missing `favcolor`?
b. How many subjects in the entire `survA` data set have no missing values at all?

11 Question 11

Consider the responses to the item "How important do you think statistics will be in your future career?" as gathered in the `statfuture` variable. There were four years prior to 2020 where the class had no missing responses on this item. Use the data from those four years to estimate the proportion of subjects who responded with the value 7 (meaning extremely important), using the approach due to Agresti and Coull.

a. Rounding to three decimal places, specify a 90% confidence interval for the true value of the proportion described in Question 11.
b. Is the sample proportion of "7" values from the 2020 survey inside the interval you reported in Question 11a?
    a. Yes, it is inside the interval
    b. No, the sample proportion is lower than all values in the interval
    c. No, the sample proportion is higher than all values in the interval

12 Question 12

Fit two linear models to predict `haircut` price on the basis of `height_in` and `english`, in each case using only those subjects with complete data on each of those three variables, as follows.

- In model **m12A**, use only `height_in` to predict `haircut`.
- In model **m12B**, use both `height_in` and `english` to predict `haircut`.

a. Specify the values of BIC for models m12A and m12B, rounded to zero decimal places.
b. According to the AIC and BIC, which of the two models (m12A or m12B) shows stronger performance?
    a. Model m12A shows stronger performance on both AIC and BIC
    b. Model m12B shows stronger performance on both AIC and BIC
    c. Model m12A is better on AIC, but m12B is better on BIC
    d. Model m12A is better on BIC, but m12B is better on AIC

13 Question 13

Using model m12B as described in Question 12, identify the standardized residuals for each subject, and identify the largest (in absolute value) of these standardized residuals.

a. Specify the `student` code for the subject with the largest (in absolute value) standardized residual in model m12B.
b. Specify the fitted and observed values of `haircut` for the subject you identified in Question 13a, in each case rounding to the nearest integer.

14 Question 14

Consider again the subject you identified by their `student` code in Question 13.

a. Rounded to three decimal places, what is the Cook's distance (using model m12B) for the subject identified in Question 13?
b. Does Cook's distance identify that subject as influential on model m12B?
    a. Yes
    b. No
    c. It is impossible to tell from the available information

15 Question 15

Now, let's create a new model, called model **m_15**, where you rerun model m12B, but this time excluding the `student` you identified in Question 13.

a. Specify the residual standard error (rounded to 1 decimal place) and the R-squared value (expressed as a proportion, and rounded to 3 decimal places) for model m_15.
b. Consider an appropriate plot of the standardized regression residuals for model m15. Is there a problem with the assumption of Normality?
    a. Yes, because there is substantial skew in the distribution of the standardized residuals
    b. Yes, because the residuals are symmetric, but there are several problematic outliers
    c. Yes, because the residuals are symmetric, but their distribution is light-tailed compared to a Normal model
    d. No, because the residuals are well-approximated by a Normal distribution

16 Question 16 (4 points)

Data from a pilot study describe the pre-operative and post-operative creatinine clearance (in ml/minute) of six patients anesthetized with a new drug combination we want to test.

Subject	Pre-operative	Post-operative
1	110	137
2	101	93
3	71	99
4	73	91
5	133	112
6	118	134

Suppose we are willing to assume that the results of this study are an appropriate pilot for a larger study we are designing that will also compare pre-operative and post-operative creatinine clearance rates. Suppose we want to maintain a two-sided 5% significance level. Suppose that an effect that is at least half the effect size seen in the pilot study will be our minimum standard for clinical importance. We also assume that the pilot's standard deviation estimates are 25% too low, so we'll multiply them by 1.25 in developing our design.

As a hint, consider the following output:

```
subj <- 1:6
pre <- c(110, 101, 71, 73, 133, 118)
post <- c(137, 93, 99, 91, 112, 134)

pilot <- tibble(subj, pre, post)

favstats(~ pre, data = pilot)
```

```
 min Q1 median  Q3 max mean       sd n missing
  71 80  105.5 116 133  101 24.81129 6       0
```

```
favstats(~ post, data = pilot)
```

```
 min   Q1 median    Q3 max mean       sd n missing
  91 94.5  105.5 128.5 137  111 20.36664 6       0
```

```
favstats(~ pre-post, data = pilot)
```

```
 min     Q1 median Q3 max mean    sd n missing
 -28 -24.75    -17  2  21  -10 19.99 6       0
```

Under the conditions specified above, you have been asked to determine the smallest number of subjects we must include in our new study to obtain 90% power to detect the minimum clinically important effect.

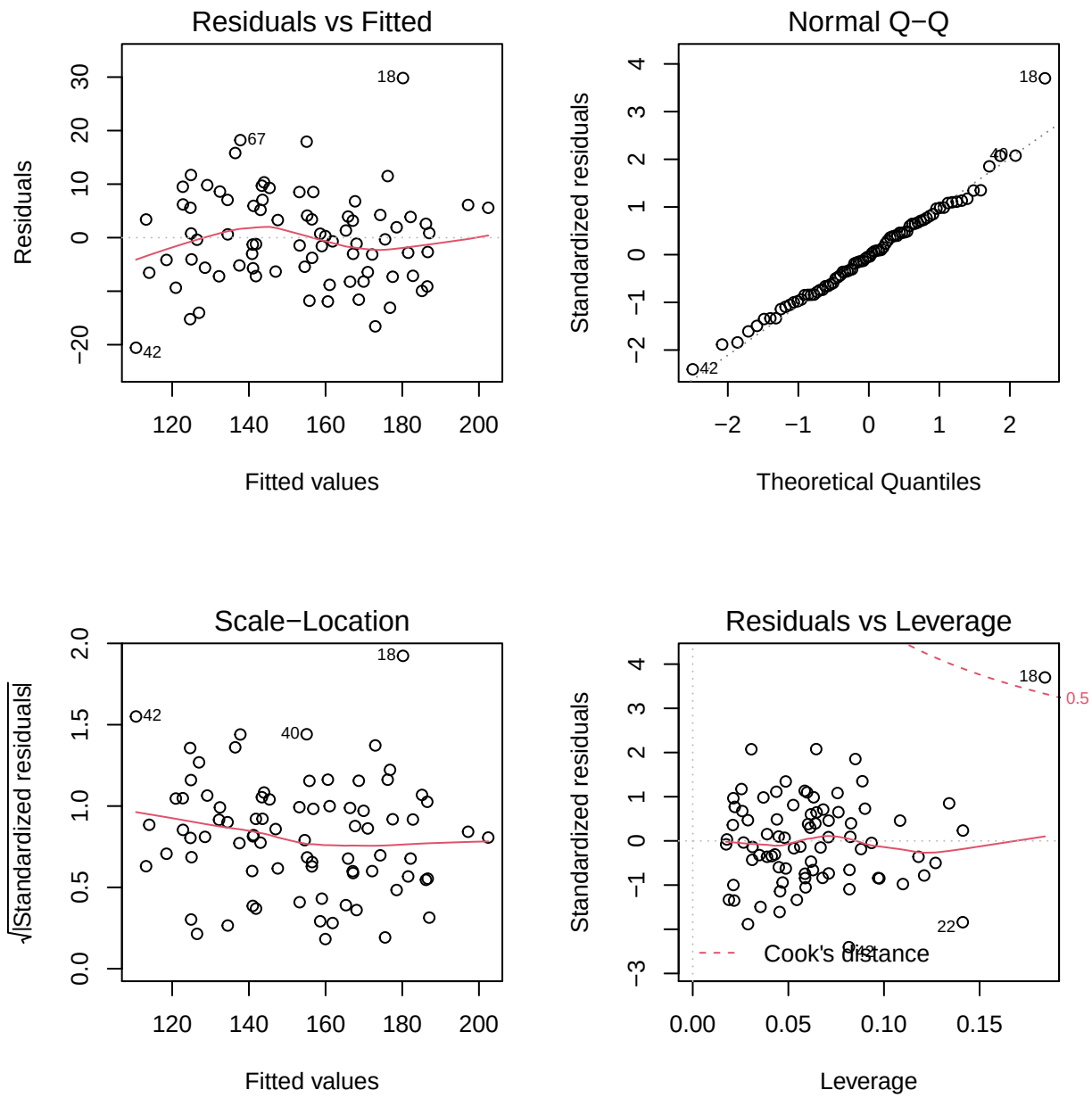
a. What is the smallest number of subjects the new study will have to include?
b. Provide your R code (a single line will suffice) to justify your response to Question 16a.

Setup for Question 17

In this question, we're going to look at three sets of regression residual plots, for three separate models, called Model 1, Model 2, and Model 3. The three models estimate different outcomes.

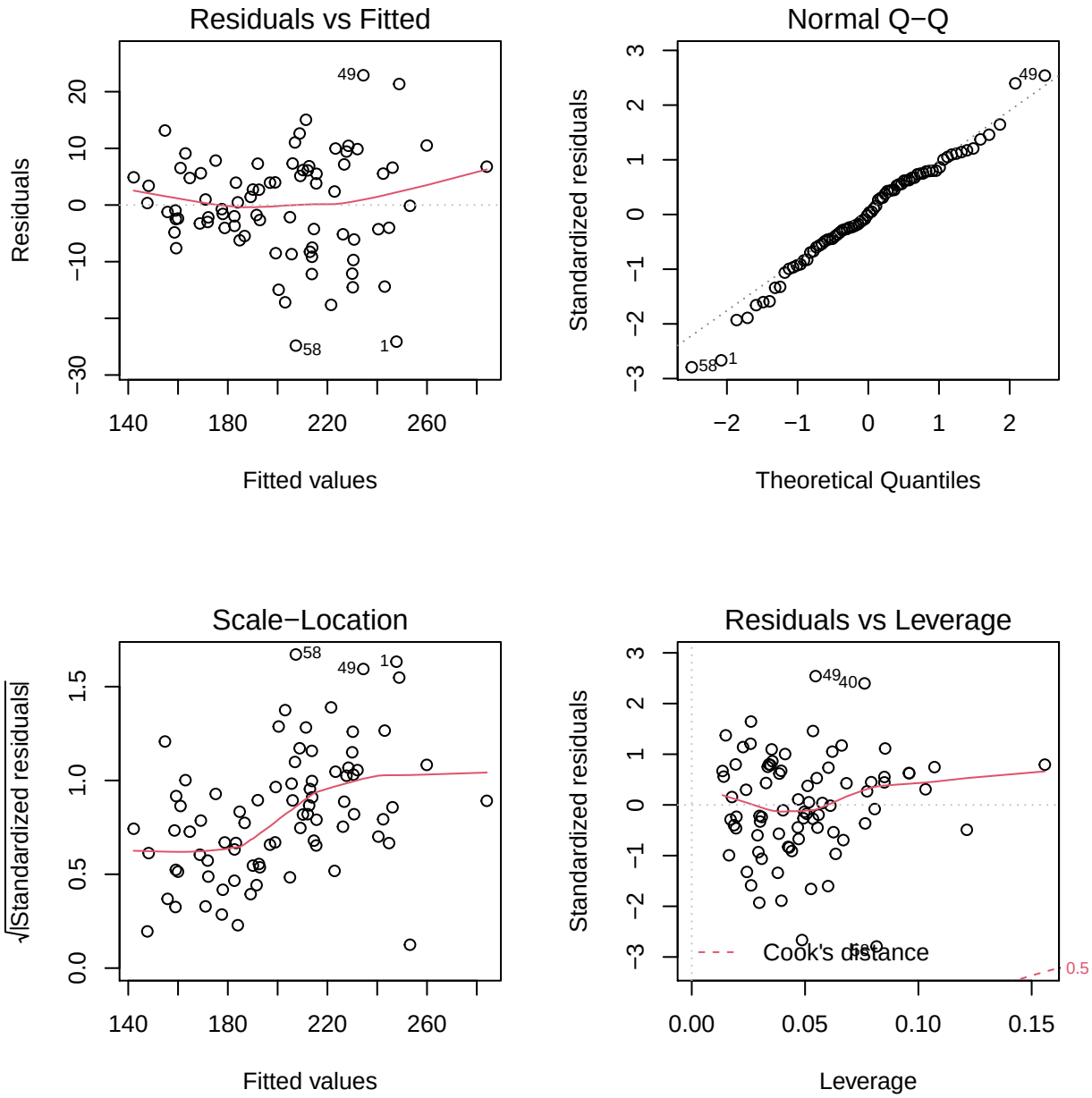
Question 17: Residual Plots - Model 1

Here are the plots for model 1, which includes 80 subjects.



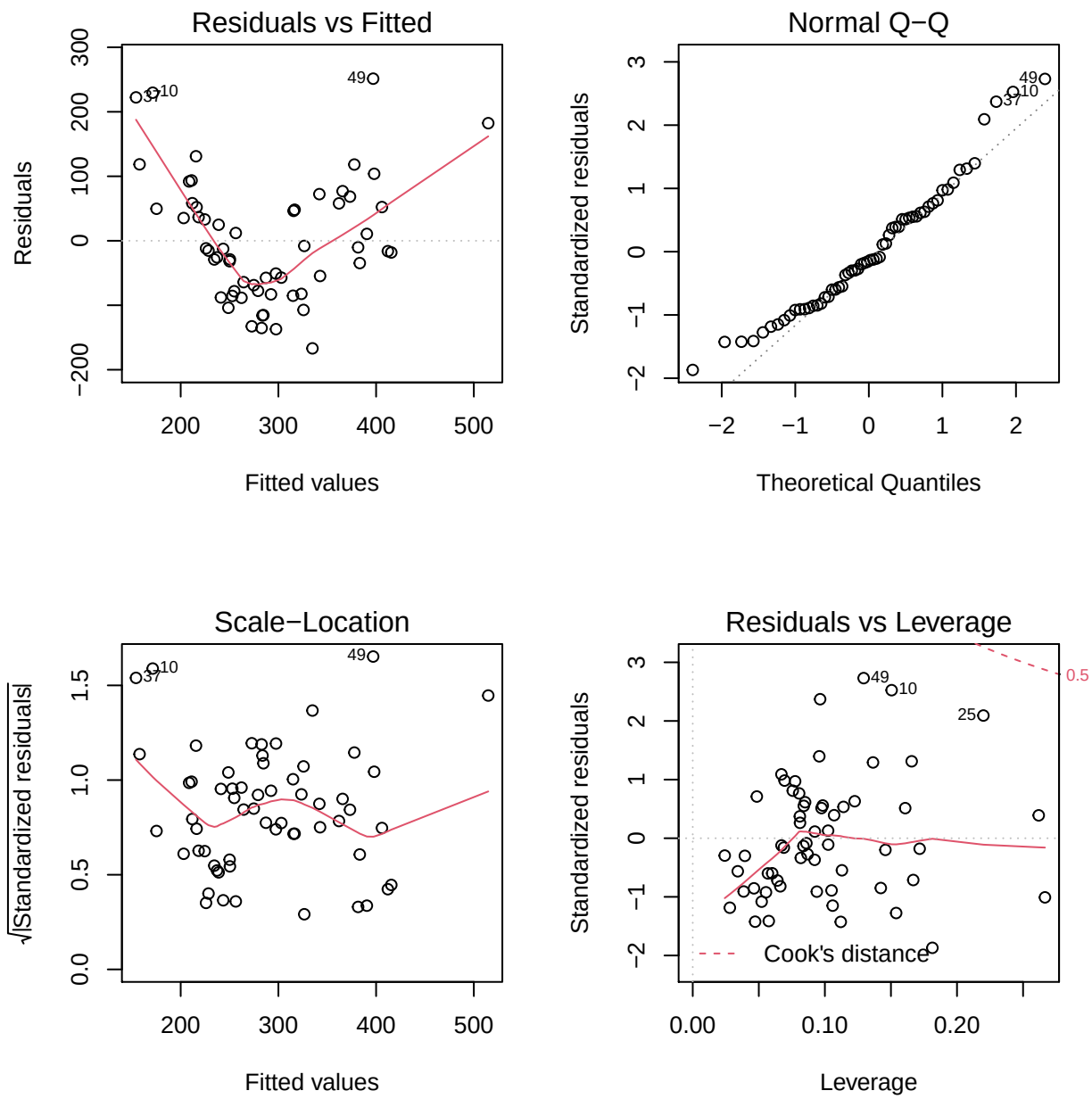
Question 17: Residual Plots - Model 2

Here are the plots for model 2, which includes 80 subjects.



Question 17: Residual Plots - Model 3

Here are the plots for model 3, which includes 60 subjects.



17 Question 17 (6 points)

For each of the three models described above, specify the most appropriate conclusion regarding its residual plots. For each model (1, 2 and 3), you are asked to select one of the following responses...

a. This model has a serious problem with the assumption of linearity
b. This model has a serious problem with the assumption of constant variance
c. This model has a serious problem with the assumption of Normality
d. There are no serious problems evident in this model's residual plots

18 Question 18

The TAs and Dr. Love prepared an analysis to repeatedly monitor accumulated results as they emerged to see if a threshold that indicated that a treatment was beneficial had been crossed. Our approach was designed to deal with a theoretical problem which David Spiegelhalter refers to in The Art of Statistics as the "Law of the Iterated Logarithm." Which of the following describes what we did in an appropriate way? (CHECK ALL RESPONSES THAT APPLY.)

a. performing a sequential probability ratio test
b. avoiding the potential for Simpson's paradox
c. using a threshold to simultaneously control Type I and Type II error
d. accounting for multiple comparisons with a Bonferroni procedure
e. None of the above

19 Question 19 (4 points)

Once a confidence interval is calculated, several design changes may be used by a researcher to make a confidence interval wider or narrower. For each of the changes listed below, indicate the impact on the width of the confidence interval.

- Rows
  a. Increase the sample size.
  b. Increase the level of confidence.
  c. Increase the standard error of the estimate.
  d. Use a bootstrap approach to estimate the CI.
- Columns
  1. CI will become wider
  2. CI will become narrower
  3. CI width will not change
  4. It is impossible to tell

20 Question 20

Suppose you have a tibble with two variables. One is a factor called Pollution with levels High, Low and Medium, arranged in that order, and the other is a quantitative outcome called Response. You want to rearrange the order of the Pollution variable so that you can then use it to identify for `ggplot2` a way to split a histogram of Response values into a series of smaller plots, each containing the histogram for subjects with a particular level of exposure (Low then Medium then High.)

Which of the pairs of `tidyverse` functions identified below would be an effective approach to use in developing such a plot?

a. `fct_reorder` and `facet_wrap`
b. `fct_relevel` and `facet_wrap`
c. `fct_collapse` and `facet_wrap`
d. `fct_reorder` and `group_by`
e. `fct_collapse` and `group_by`

21 Question 21

According to Jeff Leek in *The Elements of Data Analytic Style*, which of the following are good reasons to create graphs for data exploration? (CHECK ALL RESPONSES THAT APPLY.)

a. To understand properties of the data.
b. To inspect qualitative features of the data rather than a huge table of raw data.
c. To discover new patterns or associations.
d. To consider whether transformations may be of use.
e. To look for statistical significance without first exploring the data.

Setup for Questions 22-24

Consider the `weather_check` data frame within the `fivethirtyeight` package. We will use these data for Questions 22-24.

Suppose you want to build a table containing information from the `female`, `ck_weather` and `age` variables in that data frame. I suggest you use the following approach to place the data in the `wc` tibble, and adjust some of the coding. We'll take `female` to represent gender identification here.

Note I have also provided this code snippet to you in a file called `wc_code.R` in our Shared Google Drive in the data subfolder within the Quiz 2 folder.

```
wc <- fivethirtyeight::weather_check %>%
  select(female, ck_weather, age) %>%
  mutate(female = fct_recode(factor(female),
                             "Female" = "TRUE",
                             "Male" = "FALSE"),
         ck_weather = fct_recode(factor(ck_weather),
                            "Check" = "TRUE",
                            "No Check" = "FALSE")) %>%
  mutate(female = fct_relevel(female, "Female"),
         ck_weather = fct_relevel(ck_weather, "Check"))
```

22 Question 22

Build the specified table using your `wc` tibble. Which age group has exactly 103 respondents who identify as male who answered yes to the question "Do you typically check a daily weather report?"

a. Ages 18-29
b. Ages 30-44
c. Ages 45-59
d. Ages 60+
e. None of these.

23 Question 23

Perform an appropriate test to see if the odds ratio (for a Yes answer to "Do you typically check a daily weather report?" comparing respondents who identify as Female to those who identify as Male) is essentially consistent across age categories. What is the name of the test that you ran, and what is the conclusion? Use a 5% significance level.

a. Chi-square test on a 2x2 table with `Epi` package's `twoby2` and found a detectable association.
b. Chi-square test on a 2x2 table with `Epi` package's `twoby2` and found NO detectable association.
c. Woolf's test from the `vcd` package, and odds ratio is sufficiently consistent across age categories to let me collapse on age.
d. Woolf's test from the `vcd` package, and odds ratio is NOT sufficiently consistent across age categories to let me collapse on age.
e. None of these statements describes an appropriate test.

24 Question 24

Use the data we have been working with in the previous two questions, regardless of how you answered those questions. Suppose we want to use the Cochran-Mantel-Haenszel approach to estimate the common odds ratio across all age categories comparing the two gender groups as to whether they check the weather daily. Which of the following statements is true?

a. Females have higher odds of checking the weather, and a 95% confidence interval includes 1.
b. Females have higher odds of checking the weather, and a 95% confidence interval does not include 1.
c. Males have higher odds of checking the weather, and a 95% confidence interval includes 1.
d. Males have higher odds of checking the weather, and a 95% confidence interval does not include 1.
e. None of these statements are accurate.

25 Question 25

According to Jeff Leek in *The Elements of Data Analytic Style*, which of the following are good ideas in creating graphs you will share with other people to describe your work? (CHECK ALL RESPONSES THAT APPLY!)

a. If you have multiple plots to compare, use the same scale on the vertical axis.
b. Axis labels should be large, easy to read, in plain language.
c. Add a third dimension, perhaps with animation.
d. Include units in figure labels and legends.
e. Use color and size to help communicate information, for instance to point out confounding.

Setup for Questions 26-28

Suppose you fit four candidate regression models to predict the natural logarithm of a measure of predatory behavior in leopards. The four models are nested, in that D is a proper subset of C, which is a proper subset of B, which is a proper subset of A. Specifically, Model A contains seven predictors, Model B contains four of those seven predictors, and Model C contains three of the four Model B predictors, while Model D contains two of the predictors in Model C. You obtain the results shown below.

```
modelA <- lm(log(predatory.behavior) ~
               x1 + x2 + x3 + x4 + x5 + x6 + x7, data = leopard)
modelB <- lm(log(predatory.behavior) ~
               x1 + x2 + x3 + x4, data = leopard)
modelC <- lm(log(predatory.behavior) ~
               x1 + x2 + x3, data = leopard)
modelD <- lm(log(predatory.behavior) ~
               x1 + x2 , data = leopard)

res26 <- bind_rows(glance(modelA), glance(modelB), glance(modelC), glance(modelD)) %>%
  mutate(model_name = c("A", "B", "C", "D"))

res26 %>% select(model_name, AIC, BIC, df, df.residual, nobs)
```

```
# A tibble: 4 x 6
  model_name   AIC   BIC    df df.residual  nobs
  <chr>      <dbl> <dbl> <dbl>       <int> <int>
1 A          3763. 3801.     7         492   500
2 B          3759. 3784.     4         495   500
3 C          4161. 4182.     3         496   500
4 D          4164. 4181.     2         497   500
```

26 Question 26

Which of these models does the output suggest will be the best choice to predict the natural logarithm of the predatory behavior measure?

a. Model A
b. Model B
c. Model C
d. Model D
e. The output doesn't suggest a "best" choice

27 Question 27

The following output relates to `modelA` described in the previous question.

```
round(car::vif(modelA),3)
```

```
    x1     x2     x3     x4     x5     x6     x7
 1.009  1.122  1.138  1.010  1.010 12.269 12.257
```

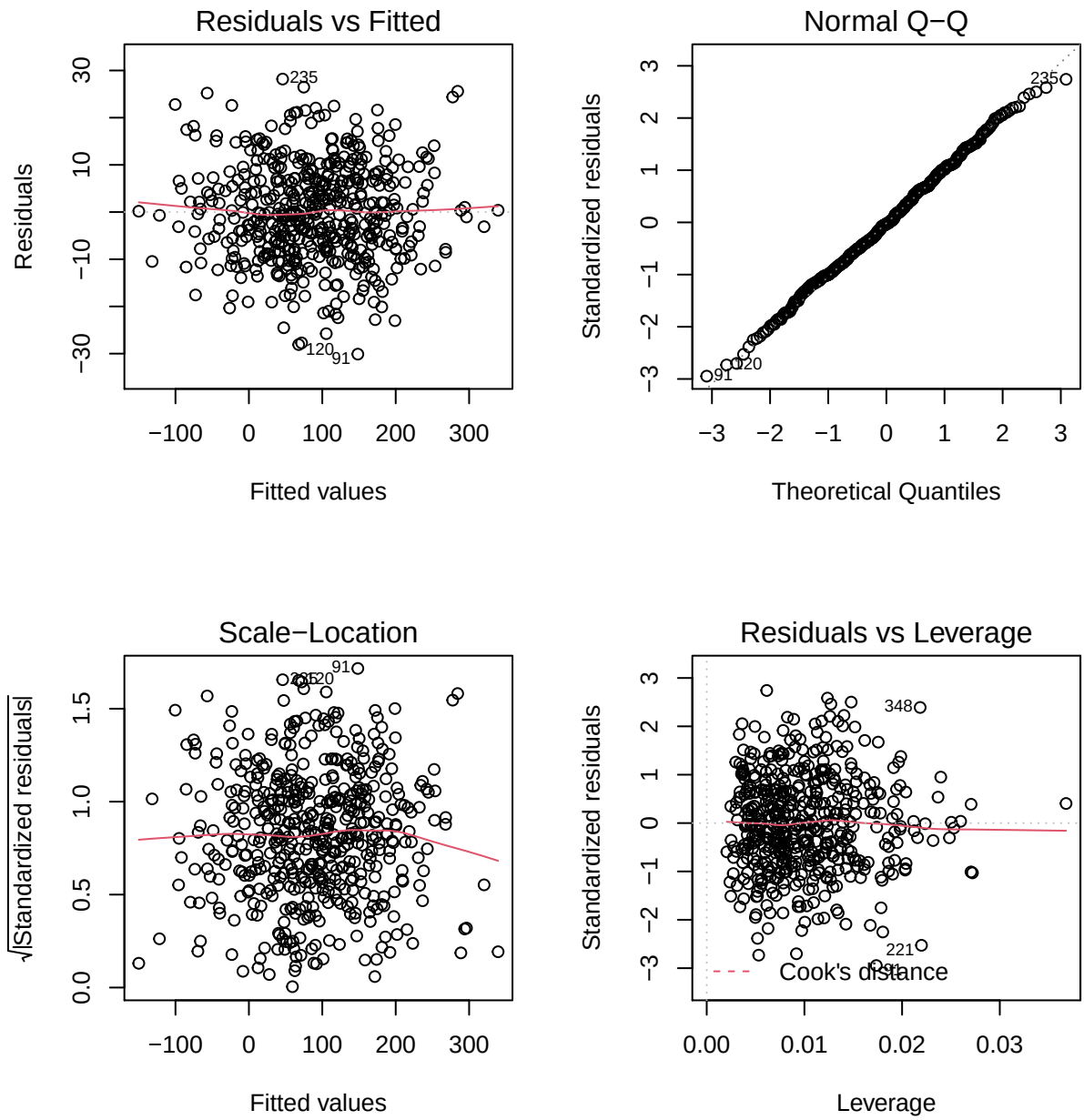
Which of the following statements is the best conclusion from the `vif` output?

a. Model A's residuals will show no problem with independence
b. Model A's residuals will show a serious problem with independence

c. Model A has no sign of meaningful collinearity
d. Model A has a serious problem with collinearity
e. None of these conclusions is appropriate

Setup for Question 28

Behold the residual plots for Model B from Question 26.



28 Question 28

Which of the following conclusions is most appropriate for model B based on the plots provided above?

a. There is a serious problem with the assumption of linearity.
b. There is a serious problem with the assumption of Normality.
c. There is a serious problem with the assumption of constant variance
d. There are no serious problems evident in these residual plots.
e. None of these conclusions are appropriate.

29 Question 29

You are comparing two regression models for the same outcome, which you built using a training sample of data. You then use each model to predict data in a test sample, that was not used to calculate the original regression equations. Which of the following summaries will be useful to you in assessing which model does a better job of out-of-sample prediction? (CHECK ALL RESPONSES THAT APPLY.)

a. The square root of the mean of the squared prediction errors
b. The mean of the absolute values of the prediction errors
c. The maximum of the absolute values of the prediction errors
d. The AIC statistic calculated in the training sample
e. None of these measures would be useful to you

30 Question 30 (8 points)

Write a clear and well-composed essay of 150 to 200 words describing an important idea from David Spiegelhalter's *The Art of Statistics* about doing statistical science well that Dr. Love didn't cover in Classes 1-24. Your essay should state the idea in your own words, and should indicate why you feel it is important.

If you quote Spiegelhalter (and we prefer that you do), specify the Chapter containing your quote. If Dr. Love discussed your idea in class, you'll lose 1 of 8 available points. If your essay is unclear, or if you miss Spiegelhalter's point, that will have a bigger impact on your score. Each Chapter in Spiegelhalter includes a summary of key points. Feel free to use these summaries to help spark ideas, but do not quote the summaries.

Use any software you like to produce a Word document or PDF submission.

These instructions are 150 words long.

31 Question 31

Have you completed the 431 class evaluation at https://webapps.case.edu/courseevals/?

a. Yes, I have.
b. Not yet, but I promise to do that before the end of the day today.
c. No, and I'm not going to promise that I will do it.

Note that Question 31 has no impact on your Quiz 2 grade, but a response of `a` or `b` will add 1 point to your class participation grade.

Affirmation

This is the final question on the Quiz. You must type in your name to answer this question before you will be permitted to submit your work in the Google Form, or save it to work on again later.

Finally, please type in your full name to affirm that you have completed this Quiz by yourself, and that you have discussed the items on this Quiz with no one other than Professor Love and the Teaching Assistants. Please type in both your first and last name.

Session Information

Here is the session information from Dr. Love's R session in which he built the Quiz, including the sketch.

```r
sessionInfo()
```

```
R version 4.0.2 (2020-06-22)
Platform: x86_64-w64-mingw32/x64 (64-bit)
Running under: Windows 10 x64 (build 19041)

Matrix products: default

locale:
[1] LC_COLLATE=English_United States.1252
[2] LC_CTYPE=English_United States.1252
[3] LC_MONETARY=English_United States.1252
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.1252

attached base packages:
[1] grid      stats     graphics  grDevices utils     datasets  methods
[8] base

other attached packages:
 [1] forcats_0.5.0     stringr_1.4.0     purrr_0.3.4       readr_1.4.0
 [5] tidyr_1.1.2       tibble_3.0.4      tidyverse_1.3.0   broom_0.7.2
 [9] Hmisc_4.4-2       Formula_1.2-4     survival_3.2-7    Epi_2.42
[13] vcd_1.4-8         patchwork_1.1.0   magrittr_2.0.1    car_3.0-10
[17] carData_3.0-4     ggrepel_0.8.2     naniar_0.6.0      mosaic_1.8.2
[21] ggridges_0.5.2    mosaicData_0.20.1 ggformula_0.9.4   ggstance_0.3.4
[25] dplyr_1.0.2       Matrix_1.2-18     ggplot2_3.3.2     lattice_0.20-41
[29] janitor_2.0.1     knitr_1.30

loaded via a namespace (and not attached):
 [1] colorspace_2.0-0    ellipsis_0.3.1      rio_0.5.16
 [4] visdat_0.5.3        leaflet_2.0.3       snakecase_0.11.0
 [7] htmlTable_2.1.0     fs_1.5.0            base64enc_0.1-3
[10] ggdendro_0.1.22     rstudioapi_0.13     farver_2.0.3
[13] fansi_0.4.1         lubridate_1.7.9.2   xml2_1.3.2
[16] splines_4.0.2       polyclip_1.10-0     jsonlite_1.7.1
[19] cluster_2.1.0       dbplyr_2.0.0        png_0.1-7
[22] ggforce_0.3.2       compiler_4.0.2      httr_1.4.2
[25] backports_1.2.0     assertthat_0.2.1    cli_2.2.0
[28] tweenr_1.0.1        htmltools_0.5.0     tools_4.0.2
```

```
[31] gtable_0.3.0        glue_1.4.2          Rcpp_1.0.5
[34] cellranger_1.1.0    vctrs_0.3.5         nlme_3.1-148
[37] crosstalk_1.1.0.1   lmtest_0.9-38       xfun_0.19
[40] rvest_0.3.6         openxlsx_4.2.3      lifecycle_0.2.0
[43] mosaicCore_0.8.0    MASS_7.3-53         zoo_1.8-8
[46] scales_1.1.1        hms_0.5.3           parallel_4.0.2
[49] RColorBrewer_1.1-2  yaml_2.2.1          curl_4.3
[52] gridExtra_2.3       rpart_4.1-15        latticeExtra_0.6-29
[55] stringi_1.5.3       checkmate_2.0.0     zip_2.1.1
[58] rlang_0.4.9         pkgconfig_2.0.3     evaluate_0.14
[61] htmlwidgets_1.5.2   cmprsk_2.2-10       tidyselect_1.1.0
[64] plyr_1.8.6          R6_2.5.0            generics_0.1.0
[67] DBI_1.1.0           pillar_1.4.7        haven_2.3.1
[70] foreign_0.8-80      withr_2.3.0         mgcv_1.8-31
[73] abind_1.4-5         nnet_7.3-14         etm_1.1.1
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[79] rmarkdown_2.5       jpeg_0.1-8.1        readxl_1.3.1
[82] data.table_1.13.2   reprex_0.3.0        digest_0.6.27
[85] numDeriv_2016.8-1.1 munsell_0.5.0
```